PROJECT EXPLAINED IN SIMPLE HINGLISH

MetroGuard

Real-World Anomaly Detection in Metro Air-Compressor Telemetry Using Bayesian Networks, RBMs and VAEs

Deep Learning II (CDT7001) | Semester VII | One-week prototype

1. Problem kya hai?

Metro train mein ek machine hoti hai: air compressor. Yeh hawa ko pressure ke saath store karti hai. Yeh compressed air train ke braking aur doosre pneumatic mechanisms ke kaam aa sakti hai.

Kabhi kabhi machine mein air leak ho jaata hai. Shuru mein kisi ko pata nahi chalta. Pressure, temperature ya motor current ki readings badal sakti hain, aur time pe na pakda jaaye toh baad mein badi kharabi ban sakti hai.

Machine pe bahut saare sensors lage hain jo time ke saath lagatar readings record karte hain. Lekin lagbhag 15 lakh readings ko insaan baithkar manually check nahi kar sakta.

Main problem

Itni saari sensor readings mein se kaise pata chale ki machine ka behaviour kab normal se alag ho raha hai?

2. Hum kya bana rahe hain?

MetroGuard ek AI-based machine-monitoring prototype hai. Isko ek smart chowkidar samjho:

- Pehle yeh machine ke normal patterns seekhega.
- Phir unusual readings milne par alert score dega: "Kuch gadbad hai!"

Simple example

Mummy ko pata hota hai tum normally kaise ho. Jis din tum chup-chup ho, unhe lag sakta hai kuch gadbad hai. Model past sensor data se patterns seekhega aur naye ya unusual pattern ko suspicious score de sakta hai.

Important distinction

Unusual behaviour detect karna aur failure hone se pehle predict karna alag cheezein hain. Hamara prototype pehle anomaly detection evaluate karega.

3. Final PS (Problem Statement)

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Real-World Anomaly Detection in Metro Air-Compressor Telemetry Using Bayesian Networks, RBMs and VAEs

Simple words: Real metro compressor ke sensor data pe alag-alag probabilistic AI models banana aur compare karna jo unusual machine behaviour pakad sakein.

Word	Matlab
Anomaly	Jo normal se alag ho
Telemetry	Sensors ki readings
Real-world	Asli machine ka asli data
BN, RBM, VAE	Teen alag probabilistic modelling approaches

4. Data kahan se aayega?

Hum UCI Machine Learning Repository ka MetroPT-3 dataset use karenge. Ismein ek operational metro train ke air compressor ke asli sensor records hain. Train kis city ya desh ki hai, UCI specify nahi karta, isliye hum location guess nahi karenge.

Official source: UCI MetroPT-3 dataset

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~15.2 lakh	15	Feb-Aug 2020	4
readings	sensor signals	approx. time period	reported air-leak intervals

File: MetroPT3(AirCompressor).csv, approximately 208 MB.

Sensors kya measure karte hain?

Sensor type	Kya batata hai
Pressure (TP2, TP3, H1, DV_pressure, Reservoirs)	System mein hawa ka pressure
Motor current	Motor kitna electrical current use kar rahi hai
Oil temperature	Oil kitna garam hai
Digital signals (8)	Kuch switches ya valves ON hain ya OFF

Data ke zaroori caveats

- Yeh real data hai, synthetic data nahi.
- Sirf 4 reported air-leak event intervals hain; har row ka perfect fault label nahi hai.
- UCI metadata mein sampling-rate ke do descriptions hain (1 Hz aur 0.1 Hz). Actual cadence timestamps se measure karenge.
- Failure table mein do events "#1" numbered hain. Event 2 ka maintenance note ("30 April") us failure (29-30 May) se pehle ka hai; report mein silently correct nahi karenge.
- Timestamps ka timezone clear nahi hai; audit mein assumption explicitly likhenge.
- Har failure window ke start/end boundaries inclusive ya exclusive hain, yeh choose karke reason report mein likhenge.

Yaad rakho

Dataset machine ki history dikhata hai, lekin 4 reported failures se har future failure predict karne ka proof nahi milta.

5. Models aur unka role

Hum paanch alag mini-projects nahi bana rahe. Ek hi compressor-monitoring problem ko syllabus ke alag probabilistic approaches se study kar rahe hain.

Unit	Model	Simple analogy	Kya karega / kyun	Limit
I	Bayesian Network	Rishton ka map	Sensor states ke probabilistic relationships. Alert ke time kaunse states saath mein unusual the, woh explore kar sakte hain.	Cause prove nahi karta.
II	Markov Network	Bina arrow wala connection map	RBM ki undirected structure, factors aur conditional independencies explain karna.	RBM ke saath explain hoga.
III	RBM	Normal patterns seekhne wala	Free-energy score ko anomaly indicator ke roop mein investigate karenge.	Binary visible units ki suitability Day 1 digital-signal audit se verify hogi.
IV	Gibbs + CD-k	Guess, check, sudhaaro	RBM training ke liye sampling aur weight updates; k = 1, 5, 10 compare karenge.	Khud implement karna project ki technical depth hai.
V	VAE	Copy / reconstruction model	Latent representation seekhkar readings reconstruct karega; reconstruction error possible anomaly signal hai.	High error apne aap fault ka proof nahi.
IV+ (bonus)	Markov state model	Machine ke moods	Motor-current operating states ke transition probabilities, agar distinct states data mein dikhein.	Optional; data aur time support kare tabhi.
-	Z-score baseline	Seedha-saadha chowkidar	Check karega ki complex models simple method se better hain ya nahi.	Sirf comparison; syllabus model nahi.

GAN, NADE, MADE: Syllabus ke theory/comparison section mein cover karenge. Inhe implemented models claim nahi karenge.

6. System kaise chalega?

1. Data audit Raw CSV load karenge. Timestamps, gaps, duplicates, digital signals, failure-window coverage aur motor current check karenge. Koi model nahi.

2. Chronological split Pehle mahine ko candidate training period maanenge, phir audit se check karenge ki koi reported failure window overlap toh nahi karta. Random shuffle nahi.

3. Model training RBM aur VAE candidate training period par train honge; BN probabilistic sensor reasoning ke liye design hoga.

4. Threshold freeze Cutoff sirf audited candidate training period se (jaise 99th percentile), test pe kabhi nahi badlega.

5. Test and compare Baaki chronological data ke scores ko reported failure intervals aur baseline ke saath compare karenge.

7. Evaluation

Candidate training period audit se confirm karne ke baad models train honge. Baaki period par test karenge, bina test failure labels se threshold tune kiye.

Metric	Sawaal
Event detection	4 reported intervals mein se kitne intervals ke saath alert align hua?
Detection delay	Alert interval ke comparison mein kitna pehle ya baad aaya?
Unconfirmed alerts/day	Reported failure windows ke bahar average kitne alerts per day aaye? Inhe unconfirmed alerts maana jayega, confirmed false alarms nahi.
Baseline comparison	RBM/VAE simple z-score se better hain ya nahi?

Fair-testing rule

Exam ke answers dekh ke padhai nahi: threshold sirf training data se decide hoga aur test par freeze rahega. Sirf 4 events hain, isliye results exploratory honge.

8. Kya claim NAHI karenge

Overclaim (galat)	Honest wording (sahi)
AI future failures predict karta hai.	Prototype unusual behaviour detect karne ki koshish karta hai.
Is sensor ne failure cause kiya.	Alert ke time yeh sensor states unusual ya associated the.
Har row ka label known hai.	Sirf 4 reported event intervals hain; row-level ground truth complete nahi.
Deployment-ready hai.	Results exploratory hain; reported events bahut kam hain.
Train Portugal ki hai.	Dataset city/country specify nahi karta; location assume nahi karenge.
BN exact wajah batata hai.	BN probabilistic reasoning ke liye hai, causal engine nahi.
Pehla mahina definitely fault-free hai.	Candidate training period hai; overlap audit se verify hoga.
Sampling rate already known hai.	Actual cadence timestamps se verify karenge.

9. Yeh project kyun? Synthetic-data idea kyun chhoda?

- Asli engineering problem: equipment ke unusual behaviour ko identify karna.
- Real operational data use karte hain; synthetic data main result nahi hai.
- Units I-V ko ek central problem se jodte hain, alag mini-projects nahi.
- RBM training, Gibbs sampling aur CD-k khud implement karenge.
- Software-only prototype: additional hardware ki zaroorat nahi; local GPU optional hai.
- Honest evaluation: chronological split, frozen threshold, baseline aur limitations.

Synthetic-data idea kyun drop kiya?

Synthetic data useful ho sakta hai, lekin is project mein uski usefulness prove karna extra research question ban jaata. MetroGuard seedhe real operational readings aur company-reported failure intervals par anomaly detection evaluate karta hai.

10. Innovation (viva ke liye)

Hum naya algorithm invent karne ka claim nahi karte. Course-project contribution yeh hai:

- RBM energy score aur VAE reconstruction error ka same real data aur same evaluation rules par comparison; optional Markov state model.
- BN se alert ke waqt sensor states ka probabilistic context.
- CD-k (k = 1, 5, 10) ka anomaly scoring par practical effect measure karna.
- Company-reported failure intervals, training-only thresholds aur simple baseline ke saath honest evaluation.

30-second viva answer

Bolne ke liye

"Ma'am, hum real metro compressor ke sensor data pe syllabus ke probabilistic models se normal patterns seekhenge aur unusual behaviour detect karenge. Hum RBM ko Gibbs sampling aur Contrastive Divergence ke saath khud implement karenge, VAE se compare karenge, aur Bayesian Network se alert ke time sensor states ke probabilistic relationships analyse karenge. Evaluation company ki reported failure intervals aur ek simple baseline ke against hoga, aur threshold sirf training period se fix karenge."

11. Viva mein aane wale sawaal

Sawaal	Jawab
Deep learning kahan hai?	RBM aur VAE neural generative models hain; RBM training khud implement karenge.
Threshold kaise decide hoga?	Sirf training data se; test par freeze rahega.
4 failures se result reliable hai?	Sample chhota hai; isliye event-level metrics report karenge aur results exploratory kahenge.
BN ne kya bataya?	Alert ke time sensor states ke probabilistic relationships; cause prove nahi karta.
Failure predict karta hai?	Nahi, abhi anomaly detection evaluate kar rahe hain. Prediction alag problem hai.
GAN kyun nahi implement kiya?	One-week scope mein core models aur valid evaluation ko priority di; GAN/NADE/MADE theory tak rahenge.

12. Tools

Tool	Use
Python	Main programming language
pandas + NumPy	CSV reading, data checks and numerical processing
Matplotlib	Sensor, failure-window and audit plots
PyTorch	RBM and VAE implementation; GPU optional
pgmpy	Bayesian Network structure and probabilistic queries
Claude Code	Implementation help after plan review and approval

13. One-week plan

Day	Kaam
Day 1	Data audit: timestamps, cadence, gaps, digital signals, failure windows, candidate split, motor current. No model.
Day 2	Bayesian Network design based on audit findings.
Days 3-4	RBM + Gibbs/CD-k implementation and reproducible experiments.
Day 5	VAE and reconstruction-based anomaly scores.
Day 6	Evaluation: frozen thresholds, event detection, delay, false alerts and baseline.
Day 7	Report and viva preparation. Markov state model optional if time is short.

Ek line mein poora project

Asli metro machine ke sensor data par AI chowkidar banate hain jo normal behaviour seekhkar unusual behaviour pakadta hai, aur imaandari se check karte hain ki kaunsa probabilistic approach sabse useful alerts deta hai.

Next: official dataset download karke Claude Code se Day 1 plan mangwao.